

LECTURE 4: BAYESIAN INFERENCE FOR VARIANCES OF NORMAL PROCESSES

Key concepts:

- Standard deviation, variance and precision
- Gamma distribution
- Gamma function
- Inverse gamma distribution
- Scaled inverse chi-squared distribution

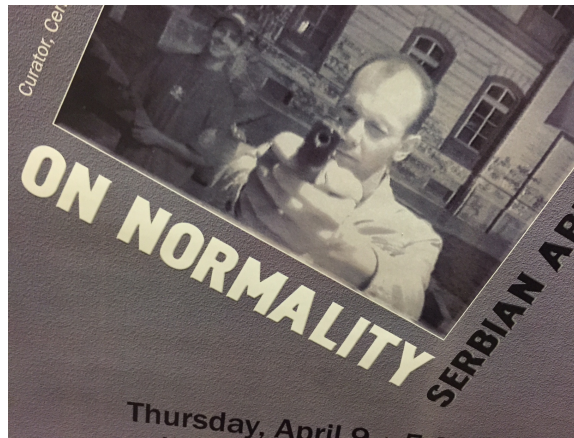


FIGURE 1. Discovered in the Encina West stairwell.

1. INTRODUCTION

Bayesian analysis of the normal distribution requires a prior distribution for the variance. In these notes, we consider the (rather unusual) problem of inference for the variance of a Normal distribution with known mean. The more realistic situation, where both mean and variance are unknown, will be treated next time. The results in Lectures 3 and 4 are of primary interest as building blocks for the normal model with unknown mean and variance.

The presentation in *BDA3* is a little confusing, so here is a road map. The conjugate prior for the variance of a Normal distribution may be specified in three *equivalent* ways:

Prior estimate based on prior-equivalent observations This is Gelman *et al.*'s preferred approach, but is non-standard. One imagines having a prior "estimate" σ_0^2 of σ^2 based upon ν_0 "prior-equivalent" observations. We use a *Scaled Inverse Chi-square* distribution for σ^2 with *degrees of freedom* ν_0 and *scale* σ_0^2 , denoted as $\sigma^2 \sim \text{Scaled Inverse Chi-square}(\nu_0, \sigma_0^2)$.

Prior shape and scale for the variance The variance parameter σ^2 is assumed to have an *Inverse Gamma* distribution with *shape* α and *scale* β . We will discuss the interpretation

of these parameters below, but the primary reason Gelman *et al.* introduce the Scaled Inverse Chi-square distribution is to ease interpretability.

Prior shape and inverse scale for the precision The precision is assumed to have a *Gamma* distribution with *shape* α and *inverse scale* β . This has been the traditional approach and makes the algebra easiest. However, precisions and inverse scale parameters are not particularly easy to interpret.

I should emphasize that there is no logical reason to prefer one approach over the other. They are all equivalent and lead to the same conclusions. Further, you may decide that three different ways of saying the same thing is overkill, but JAGS uses Gamma priors for the precision, while Stan encourages the use of the other two.

We will introduce the Gamma distribution in Section 2, which is the natural conjugate prior for the precision. In Section 3, we derive the posterior distribution for the precision using this prior. In Section 4, we introduce the inverse gamma distribution, which is the natural conjugate prior for the variance. In Section 5, we derive the posterior distribution for the variance using this prior. Finally, in Section 6, we introduce the scaled inverse chi-square distribution as an alternative parameterization of the conjugate prior for the variance.

In Section 1.6 of *BDA3*, the analysis of football point spreads provides an example where the mean is plausibly known but the variance is unknown. A histogram of the data and descriptive statistics is shown below.

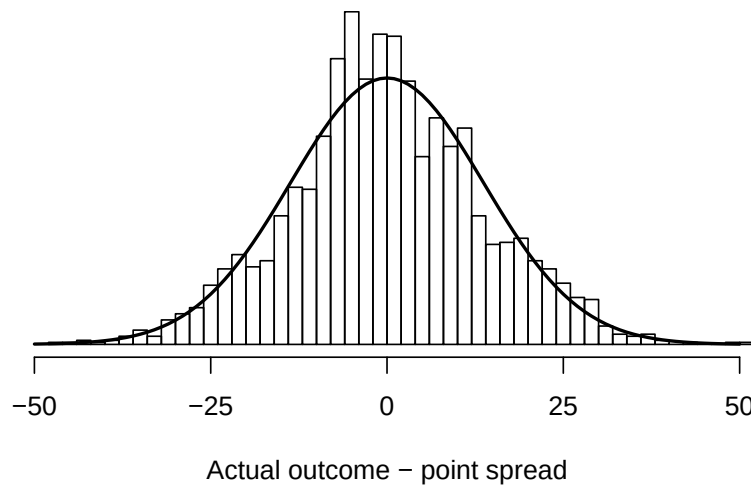


FIGURE 2. The difference between actual outcomes and point spreads in 2,240 NFL games between 1981 and 1992. The median difference is zero and the mean difference is 0.23 with a standard deviation of 13.7. The data are reasonably close to a $\text{Normal}(0, 13.7)$ distribution, which is superimposed on the histogram.

The natural conjugate prior distribution is the inverse gamma distribution. Gelman *et al.* use an alternative parameterization which they call the “scaled inverse chi-square distribution.” An equivalent formulation reparameterizes the normal distribution using its reciprocal, called the “precision,” and uses the gamma distribution for its prior. On a first encounter, this may seem rather confusing, but just remember that all three approaches—a gamma prior for the precision, an inverse gamma prior for the variance, or a scaled inverse chi-square prior for the variance—are really the same thing with different parameterizations.

2. THE GAMMA DISTRIBUTION

The *gamma distribution* is a continuous distribution with support on the positive real numbers. It has two parameters: the *shape* (usually denoted by α) and the *scale* (usually denoted by β), both of which must be positive. We write $y \sim \text{Gamma}(\alpha, \beta)$ when y has the pdf

$$p(y|\alpha, \beta) = \frac{\beta^\alpha}{\Gamma(\alpha)} y^{\alpha-1} e^{-\beta y}, \quad (y > 0, \alpha > 0, \beta > 0)$$

since

$$\int_0^\infty y^{\alpha-1} e^{-\beta y} dy = \beta^{-\alpha} \int_0^\infty x^{\alpha-1} e^{-x} dx = \frac{\Gamma(\alpha)}{\beta^\alpha}$$

where $\Gamma(\cdot)$ is the gamma function defined in Section 3.2 of Lecture 2.

In R, try plotting `dgamma(y, shape=alpha, scale=beta)` for various values of α and β to see what the distribution looks like.

Proposition 1. *If $y \sim \text{Gamma}(\alpha, \beta)$, then*

$$\mathbb{E}(y|\alpha, \beta) = \frac{\alpha}{\beta} \quad \mathbb{V}(y|\alpha, \beta) = \frac{\alpha}{\beta^2}.$$

Proof. By definition,

$$\mathbb{E}(y|\alpha, \beta) = \int_0^\infty y p(y|\alpha, \beta) dy = \frac{\beta^\alpha}{\Gamma(\alpha)} \int_0^\infty y^\alpha e^{-\beta y} dy = \frac{\beta^\alpha}{\Gamma(\alpha)} \frac{\Gamma(\alpha+1)}{\beta^{\alpha+1}} = \frac{\alpha}{\beta}.$$

The proof of the variance is similar. □

In your introductory probability course, you encountered the chi-squared and exponential distributions, which are special cases of the Gamma distribution. We explore those connections here.

2.1. Connection with the Exponential Distribution. The exponential distribution is often used for modeling positive random variables such as durations and failure times. Its pdf is given by

$$f(y|\lambda) = \lambda e^{-\lambda y} \quad (y > 0, \lambda > 0)$$

where λ is the hazard rate and $\mathbb{E}(y|\lambda) = 1/\lambda$. In this case, we write $y \sim \text{Exponential}(\lambda)$. It is evident that the $\text{Exponential}(\lambda)$ distribution is the same as the $\text{Gamma}(1, \lambda)$ distribution. From Proposition 1, we see $\mathbb{E}(y|\lambda) = 1/\lambda$ and $\mathbb{V}(y|\lambda) = 1/\lambda^2$.

2.2. Connection with the Chi-squared Distribution. The chi-squared distribution occurs frequently in statistical applications, such as tests of independence in contingency tables and the sampling distribution of sums of squares of normal random variables. The chi-squared distribution is a one-parameter family of distributions for positive random variables. The parameter is called the *degrees of freedom* and is usually denoted by ν . When a random variable y has the pdf

$$f(y|\nu) = \frac{(1/2)^{-\nu/2}}{\Gamma(\nu/2)} y^{\nu/2} e^{-y/2}, \quad (y > 0, \nu > 0)$$

we write $y \sim \chi^2(\nu)$. Matching parameters, we see that the $\chi^2(\nu)$ distribution is the same as the $\text{Gamma}(\nu/2, 1/2)$ distribution. The degrees of freedom need not be integer-valued, though when ν is an integer we have the following result.

Proposition 2. *If $z_1, \dots, z_\nu$ are ν i.i.d. standard normal random variables, then*

$$y = z_1^2 + \dots + z_\nu^2$$

has a chi-squared distribution with ν degrees of freedom.

Proof. The moment generating function of y is

$$\mathbb{E}(e^{ty}) = \mathbb{E}(\exp\{t(z_1^2 + \dots + z_\nu^2)\}) = \prod_{k=1}^{\nu} \mathbb{E}(e^{tz_k^2}) = (1 - 2t)^{-\nu/2}$$

since for $|t| < 1/2$,

$$\mathbb{E}(e^{tz_i^2}) = \int_{-\infty}^{\infty} e^{tz^2} (2\pi)^{-1/2} e^{-z^2/2} dz = (2\pi)^{-1/2} \int_{-\infty}^{\infty} e^{-(1-t)z^2} dz = (1 - 2t)^{-1/2}$$

On the other hand, if $y \sim \text{Gamma}(\nu/2, 1/2)$, then the moment generating function of y is

$$\begin{aligned} \mathbb{E}(e^{ty}) &= \int_0^{\infty} e^{ty} \frac{1/2}{\Gamma(\nu/2)} y^{\nu/2-1} e^{-y/2} dy = \frac{1/2}{\Gamma(\nu/2)} \int_0^{\infty} (y/2)^{\nu/2-1} e^{-(1-2t)y/2} dy \\ &= \frac{1}{\Gamma(\nu/2)} \int_0^{\infty} x^{\nu/2-1} e^{-(1-2t)x} dx = (1 - 2t)^{-\nu/2} \end{aligned}$$

for $|t| < 1/2$. Since the moment generating functions coincide on an open interval containing the origin, we conclude that the two distributions are the same. $\square$

If $y \sim \chi^2(\nu) = \text{Gamma}(\nu/2, 1/2)$, then

$$\mathbb{E}(y|\nu) = \frac{\nu/2}{1/2} = \nu \quad \mathbb{V}(y|\nu) = \frac{\nu/2}{(1/2)^2} = 2\nu$$

But you already knew this.

3. BAYESIAN INFERENCE FOR THE PRECISION WITH KNOWN MEAN

Suppose $y_1, \dots, y_n$ are i.i.d. $\text{Normal}(\mu, \sigma^2)$ where μ is known. Let $h = 1/\sigma^2$ denote the precision of this process. The likelihood is

$$p(y_1, \dots, y_n|h) = (2\pi)^{-n/2} h^{-n/2} \exp\left\{-\frac{h}{2} S(\mu)\right\}$$

where $S(\mu) = \sum_{i=1}^n (y_i - \mu)^2$ is sample sum of squares around μ .

As a function of the precision h , the form of the likelihood function is

$$p(y_1, \dots, y_n | h) \propto h^{A-1} e^{-Bh}$$

which is proportional to a Gamma pdf with shape $A = n/2 + 1$ and scale $B = S(\mu)/2$. This suggests the natural conjugate prior $h \sim \text{Gamma}(\alpha, \beta)$,

$$p(h) = \frac{\beta^\alpha}{\Gamma(\alpha)} h^{\alpha-1} e^{-\beta h}, \quad (h > 0)$$

where the prior mean is $\mathbb{E}(h) = \alpha/\beta$ and the prior variance is $\mathbb{V}(h) = \alpha/\beta^2$.

With this model and prior, the posterior is

$$\begin{aligned} p(h | y_1, \dots, y_n) &\propto p(h) p(y_1, \dots, y_n | h) \\ &\propto (h^{\alpha-1} e^{-\beta h}) (h^{n/2} e^{-hS(\mu)/2}) \\ &= h^{\alpha+n/2-1} e^{-(\beta+S(\mu)/2)h}, \end{aligned}$$

which is proportional to a $\text{Gamma}(\alpha + n/2, \beta + S(\mu)/2)$ pdf. The prior mean was

$$\mathbb{E}(h) = \frac{\alpha}{\beta},$$

while the posterior mean is

$$\mathbb{E}(h | y_1, \dots, y_n) = \frac{\alpha + n/2}{\beta + S(\mu)/2}.$$

The posterior mean is greater than the prior mean whenever $n/S(0) > \alpha/\beta$. We can think of $n/S(\mu)$ as the sample precision and we adjust the posterior mean of the precision $\mathbb{E}(h | y_1, \dots, y_n)$ upward whenever the sample precision exceeds its prior expectation $\mathbb{E}(h) = \alpha/\beta$.

4. THE INVERSE GAMMA DISTRIBUTION

If $x \sim \text{Gamma}(\alpha, \beta)$ and $y = 1/x$, then the pdf of y is

$$p_y(y) = p_x(1/y) \left| \frac{dx}{dy} \right| = \frac{\beta^\alpha}{\Gamma(\alpha)} (1/y)^{\alpha-1} e^{-\beta/y} \left| -\frac{1}{y^2} \right| = \frac{\beta^\alpha}{\Gamma(\alpha)} y^{-\alpha-1} e^{-\beta/y}$$

for $y > 0$. In this case, we say that y has an *inverse gamma distribution*. The parameters α and β must be positive. If $y \sim \text{Inverse Gamma}(\alpha, \beta)$ with $\alpha > 1$, then

$$\mathbb{E}(y | \alpha, \beta) = \frac{\beta}{\alpha - 1}.$$

If, $\alpha > 2$, then

$$\mathbb{V}(y | \alpha, \beta) = \frac{\beta^2}{(\alpha - 1)^2(\alpha - 2)}.$$

We can simulate n draws from the inverse gamma distribution in R using `1/rgamma(n, shape=alpha, rate=beta)`.

5. BAYESIAN INFERENCE FOR THE VARIANCE WITH KNOWN MEAN

Next, we reformulate the Bayesian analysis with the variance instead of the precision. We now start with the prior

$$\sigma^2 \sim \text{Inverse Gamma}(\alpha, \beta)$$

and likelihood

$$p(y_1, \dots, y_n | \sigma^2) = (2\pi)^{-n/2} (\sigma^2)^{-n/2} \exp \left\{ -\frac{S(\mu)}{2\sigma^2} \right\}.$$

The prior pdf is

$$p(\sigma^2) \propto (\sigma^2)^{-\alpha-1} e^{-\beta/\sigma^2}$$

so the posterior pdf is

$$\begin{aligned} p(\sigma^2 | y_1, \dots, y_n) &\propto p(\sigma^2) p(y_1, \dots, y_n | \sigma^2) \\ &\propto (\sigma^2)^{-\alpha-1} e^{-\beta/\sigma^2} (\sigma^2)^{-n/2} \exp \left\{ -\frac{S(\mu)}{2\sigma^2} \right\} \\ &\propto (\sigma^2)^{-\alpha+n/2-1} \exp \left\{ -\frac{\beta + S(\mu)/2}{\sigma^2} \right\}, \end{aligned}$$

which is proportional to an Inverse Gamma($\alpha + n/2, \beta + S(\mu)/2$) distribution. The posterior moments of σ^2 are

$$\mathbb{E}(\sigma^2 | y_1, \dots, y_n) = \frac{\beta + S(\mu)/2}{\alpha + n/2 - 1} \quad \mathbb{V}(\sigma^2 | y_1, \dots, y_n) = \frac{(\beta + S(0)/2)^2}{(\alpha + n/2 - 1)^2(\alpha + n/2 - 2)}$$

6. SCALED INVERSE CHI-SQUARED PRIORS

The parameters of the gamma and inverse gamma distributions are not the most natural ones for Bayesian analysis of the variance of a normal process. Gelman *et al.*, *BDA3*, define the *scaled inverse Chi-squared* distribution with parameters ν (degrees of freedom) and s^2 (scale). This is just a reparameterization of the inverse gamma distribution with $\alpha = \nu/2$ and $\beta = \nu s^2/2$:

$$\begin{aligned} p(\sigma^2) &= \frac{(\nu s^2/2)^{\nu/2}}{\Gamma(\nu/2)} (\sigma^2)^{-\nu/2-1} e^{-\nu s^2/(2\sigma^2)} \\ &= \frac{(\nu/2)^{\nu/2}}{\Gamma(\nu/2)} s^\nu (\sigma^2)^{-(\nu/2+1)} e^{-\nu s^2/(2\sigma^2)}, \end{aligned}$$

which is the version given in Table A.1 in *BDA3*. The mean and variance of the scaled inverse chi-squared distribution are

$$\begin{aligned} \mathbb{E}(\sigma^2 | \nu, s^2) &= \frac{\nu s^2/2}{\nu/2 - 1} = \frac{\nu}{\nu - 2} s^2 \\ \mathbb{V}(\sigma^2 | \nu, s^2) &= \frac{(\nu s^2/2)^2}{(\nu/2 - 1)^2(\nu/2 - 2)} = \frac{2\nu^2}{(\nu - 2)^2(\nu - 4)} s^4 \end{aligned}$$

Thus, s^2 is (approximately) the prior mean of σ^2 and ν is (approximately) the prior equivalent number of observations.¹

¹The information for σ^2 in $y_1, \dots, y_n$ is $2n/\sigma^4$.

Now, suppose we start with the prior distribution $\sigma^2 \sim \text{Scaled Inverse Chi-squared}(\nu_0, s_0^2)$, which has kernel

$$p(\sigma^2) \propto (\sigma^2)^{-(\nu_0/2+1)} \exp \left\{ -\frac{\nu_0 s_0^2}{2\sigma^2} \right\}.$$

The kernel of the posterior distribution is

$$\begin{aligned} p(\sigma^2 | y_1, \dots, y_n) &\propto p(\sigma^2) p(y_1, \dots, y_n | \sigma^2) \\ &\propto (\sigma^2)^{-(\nu_0/2+1)} \exp \left\{ -\frac{\nu_0 s_0^2}{2\sigma^2} \right\} (\sigma^2)^{-n/2} \exp \left\{ -\frac{S(\mu)}{2\sigma^2} \right\} \\ &\propto (\sigma^2)^{-[(\nu_0+n)/2+1]} \exp \left\{ -\frac{\nu_0 s_0^2 + nv}{2\sigma^2} \right\} \end{aligned}$$

where $v = S(\mu)/n$. This is a Scaled Inverse Chi-square(ν_n, s_n^2) distribution where

$$\begin{aligned} \nu_1 &= \nu_0 + n \\ s_n^2 &= \frac{\nu_0 s_0^2 + nv}{\nu_0 + n} \end{aligned}$$

The posterior degrees of freedom ν_n adds n observations to the ν_0 prior degrees of freedom. The posterior scale parameter s_n^2 is a weighted average of the prior scale parameter s_0^2 and $v = S(\mu)/n$ (the mean sum of squares in the sample around μ).

TABLE 1. Relationship Between Priors

Distribution	Variable	Prior Parameters	Prior Mean	Posterior Mean
Gamma	Precision	Shape = α , Rate = β	$\mathbb{E}(h) = \frac{\alpha}{\beta}$	$\frac{\alpha + n/2}{\beta + S(\mu)/2}$
Inverse Gamma	Variance	Shape = α , Rate = β	$\mathbb{E}(\sigma^2) = \frac{\beta}{\alpha - 1}$	$\frac{\beta + S(\mu)/2}{\alpha + n/2 - 1}$
Scaled Inverse Chi-square	Variance	d.f. = ν_0 , Scale = s_0^2	$\mathbb{E}(\sigma^2) = \frac{\nu_0}{\nu_0 - 2} s_0^2$	$\frac{\nu_0 s_0^2 + nv}{\nu_0 + n}$